Machine Learning-Based

Predicting At-Risk Students from Social and Behavioral Factors

A Machine Learning Approach to Early Academic Intervention

IC-ML Capstone Project Proposal

Group 4

Khalil Badal
Amiel Elestin Cruz
Sherman Lloyd Ko
Elijah David Lopez
Klein Isshi Trangia
Arc Joseph Villarmil

University of Santo Tomas

Why Academic Success Matters

- Academic performance strongly shapes a student's future opportunities.
- Yet in every class, some students quietly fall behind.
- Support depends on noticing them in time.

But how do we know who is struggling - before it is too late?

The Problem: Risk Is Seen Too Late

- The clearest signal of academic trouble is the final grade.
- By the time it arrives, the term is over and intervention is no longer possible.
- A single teacher handles many students and cannot track each one's risk.

The Cost of Late Detection

- Missed window to provide tutoring, counseling, or extra support.
- Grade retention - repeating the subject or the year.
- Higher likelihood of disengagement and dropping out.

How Is It Handled Now?

- Support is mostly reactive - it comes after a student already fails.
- Identification relies on manual observation, limited by class size.
- There is no systematic, early signal of who is at risk.

How can machine learning help identify at-risk students early?

Existing Machine Learning Approaches

- Prior studies predict student performance using classifiers such as decision trees, SVM, and neural networks (Cortez & Silva; Yagci; Shahiri et al.).
- They show that student outcomes are learnable from demographic, social, and behavioral data.
- Gap: many rely on prior or intermediate grades as inputs - which only become available too late for early intervention.

Problem Statement

Academic risk is typically identified only at the end of the term, from grades that appear too late for meaningful intervention. There is a need to identify students at risk of failing early - using only information available before final assessment - so that support can be given while it still matters.

Motivation

This project aims to build a data-driven early-warning tool that flags students at risk of failing from early, non-grade factors. By predicting risk before finals, the model can help teachers and guidance staff prioritize support for the students who need it most.

Dataset: Rationale & Overview

UCI Student Performance Dataset (Mathematics)

395	33	Binary
Students	Attributes	Pass / Fail

- Roughly one in three students fail - the classes are imbalanced.
- Chosen because it is public, reliable, and rich in social and behavioral factors.
- A widely used benchmark for student-performance prediction.
- Source: UCI Machine Learning Repository.

Dataset: Key Terminology

The 33 attributes fall into four groups:

- Demographic - age, sex, home address, family size
- Family background - parents' education and jobs, guardian
- Academic history - past failures, weekly study time, school and family support
- Behavioral / social - absences, free time, going out, alcohol use, activities

TARGET VARIABLE

PASS when final grade G3 is 10 or above | AT RISK (fail) when G3 is below 10

The three grades - G1, G2, and G3 - are explained next.

Understanding the Grades: G1, G2, G3

G1

First Period

G2

Second Period

G3

Final Grade

Like our Prelim and Midterm grades - they hint at the outcome, but the Final Grade is the real result.

- All three are graded on a 0-20 scale, where 10 and above is passing.
- G1 and G2 are interim grades earned during the year; G3 is the official grade that decides passing.
- Earlier grades only hint at the result - a student can still recover or collapse.
- In the data: 24 students recovered from a failing 2nd period, while 8 who were passing still failed.

The Role of G1 and G2

Scenario 1 - With G1 & G2

- Uses all attributes, including the two interim grades.
- Expected to be highly accurate - G2 alone matches the final 92% of the time.
- Reflects what a teacher already knows by mid-year.

Scenario 2 - Without G1 & G2

- Uses only demographic, social, and behavioral factors.
- A harder, earlier prediction - before any grade exists.
- Shows how far risk can be flagged from stable factors alone.

We compare both - to measure how much the interim grades add, and how early we can still predict.

General Objective

Main Objective

To develop and evaluate a machine learning-based classification solution that identifies secondary-school students at risk of failing the final grade, using their academic, social, and behavioral attributes, in order to support timely intervention.

Specific Objectives

1. Train and compare classification models (KNN, Logistic Regression, SVM) to identify students at risk of failing the final grade.
2. Compare performance with and without the interim grades (G1, G2) to measure their contribution and the feasibility of earlier prediction.
3. Apply feature selection and dimensionality reduction to find which factors most influence the risk of failing.
4. Evaluate and compare the models using appropriate metrics and validation, prioritizing recall and F1-score on the at-risk group.
5. Deliver a documented, reproducible Python implementation as a prototype early-warning tool.

Proposed Methodology

1. Dataset Acquisition
2. Data Preprocessing
3. Model Development
4. Feature Selection & PCA
5. Model Evaluation
6. Prototype Demo

Implemented in Python (Spyder) with pandas, scikit-learn, and Matplotlib / Seaborn.

Proposed Models & Evaluation

Logistic Regression	K-Nearest Neighbors	Support Vector Machine
Linear probabilistic baseline	Classifies by most similar students	Maximum-margin boundary, kernels

Evaluation Metrics (focus on the minority at-risk class)

- Recall - catch the students who actually fail (most important).
- Precision - limit false alarms among those flagged.
- F1-score - balance recall and precision in one number.
- Confusion matrix & ROC-AUC - full picture beyond accuracy.
- Validation - appropriate cross-validation for robust results.

Expected Output & Contributions

- A complete, documented Python pipeline - from preprocessing to evaluation.
- A comparison of three classifiers for at-risk detection.
- A measure of how much the interim grades (G1, G2) improve prediction - and how early risk can be flagged without them.
- A ranked set of the factors most associated with academic risk (e.g., study time, past failures, absences).
- A validated early-warning prototype that supports timely intervention.

Projected Timeline

Phase 1	**Weeks 1-2**	Dataset acquisition, cleaning, and exploratory data analysis
Phase 2	**Weeks 3-4**	Preprocessing, encoding, and deriving the pass / fail target
Phase 3	**Weeks 5-6**	Development and training of the classification models
Phase 4	**Weeks 7-8**	Feature selection, PCA, and hyperparameter tuning
Phase 5	**Weeks 9-10**	Evaluation, cross-validation, and results visualization
Phase 6	**Weeks 11-12**	Final paper, documentation, and presentation



Thank You!

Group 4 - IC-ML Machine Learning Capstone